Explicit Constructions of Generalized Intransitive Dice

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Abstract

This paper presents an explicit construction of generalized intransitive dice. A general expression for a set of intransitive dice of any given size is provided, and the probability of one die rolling higher than another within the set is explicitly computed. Using this expression, the non-transitive property of these sets is rigorously demonstrated, alongside an analysis of their inherent symmetries.

1 Introduction

In probability theory, a set of dice is considered intransitive if it contains dice such that, for any given die in the set, there exists another die that rolls a higher number with a probability strictly greater than one-half. Consequently, the relation "die A beats die B" (denoted as $A > B$) is not transitive. Constructing such sets is generally non-trivial, as their underlying structure defies intuitive expectations. Intransitive dice are not restricted to the standard six faces, and the number of dice in a set may vary, provided there are at least three dice to form a non-degenerate cycle.

It is instructive to note that any set of intransitive dice can be modeled as a *tournament*—a complete directed graph where vertices represent the dice, and a directed edge $A \to B$ indicates that A beats B with a probability greater than $1/2$ [5]. Thus, the study of intransitive dice is deeply intertwined with the combinatorial properties of tournaments.

Historically, various specific configurations of intransitive dice have been discovered, varying in the number of dice and the number of faces. Several notable examples are detailed below, where the face values of each die are represented as row vectors.

Efron's Dice Efron's dice consist of four six-sided dice. Notably, face values are repeated within individual dice, and a trivial die (D_2) with identical faces is included. However, no single face value is shared between any two dice. It can be verified that $\mathbb{P}[D_1 < D_2] > \frac{1}{2}$, $\mathbb{P}[D_2 < D_3] > \frac{1}{2}$, $\mathbb{P}[D_3 < D_4] > \frac{1}{2}$, and $\mathbb{P}[D_4 < D_1] > \frac{1}{2}$. Therefore, the cyclic relation $D_1 > D_4 > D_3 > D_2 > D_1$ holds.

$$\begin{array}{rl} D_1: & [\; 2 \;\; 2 \;\; 2 \;\; 2 \;\; 6 \;\; 6 \;] \\ D_2: & [\; 3 \;\; 3 \;\; 3 \;\; 3 \;\; 3 \;\; 3 \;] \\ D_3: & [\; 4 \;\; 4 \;\; 4 \;\; 4 \;\; 0 \;\; 0 \;] \\ D_4: & [\; 5 \;\; 5 \;\; 5 \;\; 1 \;\; 1 \;\; 1 \;] \end{array} \tag{1}$$

Miwin's Dice Miwin's dice comprise three six-sided dice. This set exhibits a remarkable symmetry in the winning probabilities: $\mathbb{P}[D_1 < D_2] = \mathbb{P}[D_2 < D_3] = \mathbb{P}[D_3 < D_1]$. Unlike Efron's dice, no face value is repeated within a single die, although some values are shared across different dice. The non-transitive relation is $D_1 > D_3 > D_2 > D_1$.

$$\begin{array}{rl} D_1: & [\; 1 \;\; 2 \;\; 5 \;\; 6 \;\; 7 \;\; 9 \;] \\ D_2: & [\; 1 \;\; 3 \;\; 4 \;\; 5 \;\; 8 \;\; 9 \;] \\ D_3: & [\; 2 \;\; 3 \;\; 4 \;\; 6 \;\; 7 \;\; 8 \;] \end{array} \tag{2}$$

Oskar's Dice Oskar van Deventer introduced a set of seven intransitive dice. The original version employs standard six-sided dice, where each of the following face values is repeated twice per die.

$$\begin{array}{rl} D_1: & [\; 2 \;\; 14 \;\; 17 \;] \\ D_2: & [\; 7 \;\; 10 \;\; 16 \;] \\ D_3: & [\; 5 \;\; 13 \;\; 15 \;] \\ D_4: & [\; 3 \;\; 9 \;\; 21 \;] \\ D_5: & [\; 1 \;\; 12 \;\; 20 \;] \\ D_6: & [\; 6 \;\; 8 \;\; 19 \;] \\ D_7: & [\; 4 \;\; 11 \;\; 18 \;] \end{array} \tag{3}$$

This set exhibits complex dominance relations: $D_1 > \{D_2, D_3, D_4\}$, $D_2 > \{D_3, D_4, D_6\}$, $D_3 > \{D_4, D_5, D_7\}$, $D_4 > \{D_1, D_5, D_6\}$, $D_5 > \{D_2, D_6, D_7\}$, $D_6 > \{D_1, D_3, D_7\}$, and $D_7 > \{D_1, D_2, D_4\}$.

Grime's Dice Dr. James Grime discovered a set of five intransitive dice. Similar to Efron's dice, faces are repeated within each die, but no face values are shared among the dice.

$$\begin{array}{rl} D_4: & [\ 4\ \ 4\ \ 4\ \ 4\ \ 4\ \ 9\ ] \\ D_5: & [\ 3\ \ 3\ \ 3\ \ 3\ \ 8\ \ 8\ ] \\ D_1: & [\ 2\ \ 2\ \ 2\ \ 7\ \ 7\ \ 7\ ] \\ D_2: & [\ 1\ \ 1\ \ 6\ \ 6\ \ 6\ \ 6\ ] \\ D_3: & [\ 0\ \ 5\ \ 5\ \ 5\ \ 5\ \ 5\ ] \end{array} \tag{4}$$

Tetrahedra Intransitive sets utilizing dice with different numbers of faces have also been formulated. For instance, the following set uses four-sided dice (tetrahedra).

$$\begin{array}{rl} D_1: & [\ 1\ \ 4\ \ 7\ \ 7\ ] \\ D_2: & [\ 2\ \ 6\ \ 6\ \ 6\ ] \\ D_3: & [\ 3\ \ 5\ \ 5\ \ 8\ ] \end{array} \tag{5}$$

A practical application of these sets is to gain an advantage in a betting game where players sequentially choose a die, wagering on who will roll the highest number. In such a scenario, the last player to choose (the "hero") can always select a die that statistically dominates all previously chosen dice. Efron's and Miwin's dice are suitable for two-player games, whereas Oskar's and Grime's dice can accommodate three players. This paper restricts its focus to two-player tournaments.

In this work, we provide an explicit formula for a generalized set of intransitive dice of arbitrary size N, where each die possesses N faces. These sets facilitate two-player *High Roller* tournaments. While previous literature offers extensive constructions capable of modeling arbitrary tournaments—providing explicit methods and bounds on the required number of faces [1]—our contribution emphasizes a more specific, streamlined construction that elegantly illustrates the non-transitive phenomenon in a simplified context.

2 Definitions

This section establishes the notation employed throughout the paper and formalizes the concepts and hypotheses necessary for the subsequent analysis.

2.1 Set of Dice

For a given integer $N \geq 3$, we define a set of dice, indexed by $n \in \{1, 2, \ldots, N\}$, as a collection of uniform discrete random variables $\{D_n\}_{n=1}^{N}$. Each variable takes values in a finite set $V_n = \{v_{n,1}, v_{n,2}, \ldots, v_{n,M}\} \subset \mathbb{N}$. We assume fair dice, meaning that the probability of any specific outcome is uniform:

$\mathbb{P}[D_n = v_{n,j}] = \frac{1}{M}$ for $j = 1, 2, \ldots, M$. In the context of this paper, we consider sets of N fair dice, each possessing M faces.

2.2 Intransitive Properties

We formalize two distinct classifications of the intransitive property.

2.2.1 Weak Intransitive Property

A set of dice $\{D_n\}_{n=1}^{N}$ is defined as *weakly intransitive* if, for every $m \in \{1, 2, \ldots, N\}$, there exists an $n \in \{1, 2, \ldots, N\}$ such that:

$$\mathbb{P}[D_m < D_n] > \frac{1}{2}. \tag{6}$$

This implies that no single die in the set is optimal; every die is statistically dominated by at least one other die.

2.2.2 Strong Intransitive Property

A set of dice $\{D_n\}_{n=1}^{N}$ is defined as *strongly intransitive* if, for all $m, n \in \{1, 2, \ldots, N\}$, the following implication holds:

$$(n - m) \,\%\, (N) < N/2 \implies \mathbb{P}[D_m < D_n] > \frac{1}{2}. \tag{7}$$

This indicates a highly structured dominance hierarchy where each die is beaten by exactly $\lfloor N/2 \rfloor$ other dice in a cyclic manner.

Visually, if the dice are arranged in a circle, each die dominates the dice located on one side of it and is dominated by the dice on the opposite side. This conceptualization is illustrated in Figure 1.

3 Generalized Set of Intransitive Dice

For any given $N \geq 3$, we propose the following generalized formula for the face values of a set of intransitive dice:

$$v_{n,j} = (j - 1)N + (n - j) \,\%\, (N) + 1. \tag{8}$$

This formula generates a matrix of face values structured as follows:

$$\begin{array}{lclllllll}
D_1 & : & [& 1 & N + N & 2N + (N-1) & \cdots & (N-2)N + 3 & (N-1)N + 2 &] \\
D_2 & : & [& 2 & N + 1 & 2N + N & \cdots & (N-2)N + 4 & (N-1)N + 3 &] \\
D_3 & : & [& 3 & N + 2 & 2N + 1 & \cdots & (N-2)N + 5 & (N-1)N + 4 &] \\
\cdots & & & & & & \cdots & & \cdots & \\
D_{N-2} & : & [& N - 2 & N + (N-3) & 2N + (N-4) & \cdots & (N-2)N + N & (N-1)N + (N-1) &] \\
D_{N-1} & : & [& N - 1 & N + (N-2) & 3N + (N-3) & \cdots & (N-2)N + 1 & (N-1)N + N &] \\
D_N & : & [& N & N + (N-1) & 3N + (N-2) & \cdots & (N-2)N + 2 & (N-1)N + 1 &].
\end{array} \tag{9}$$

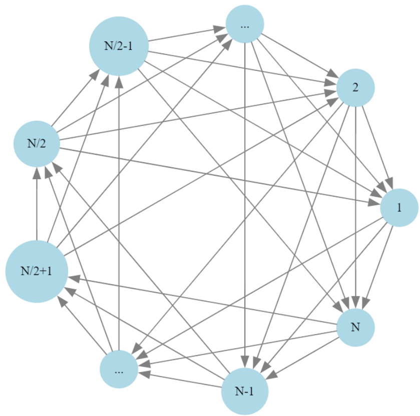

Figure 1: Cyclic dominance diagram of strongly intransitive dice.

A more intuitive interpretation is that the face values are constructed by adding a cyclically permuted vector $[1, 2, \ldots, N]$ to a base column vector:

$$\begin{array}{lllllllll} D_1 & : & \bar{v} + [& 1^* & N & N-1 & \cdots & 3 & 2 &] \\ D_2 & : & \bar{v} + [& 2 & 1^* & N & \cdots & 4 & 3 &] \\ D_3 & : & \bar{v} + [& 3 & 2 & 1^* & \cdots & 5 & 4 &] \\ \cdots & & & & & & \cdots & & \cdots & \\ D_{N-2} & : & \bar{v} + [& N-2 & N-3 & N-4 & \cdots & N & N-1 &] \\ D_{N-1} & : & \bar{v} + [& N-1 & N-2 & N-3 & \cdots & 1^* & N &] \\ D_N & : & \bar{v} + [& N & N-1 & N-2 & \cdots & 2 & 1^* &], \end{array} \tag{10}$$

where $\bar{v} = [0, N, 2N, \ldots, (N-1)N]$.

In this specific construction, a shift of a single position is utilized. However, any generator of the cyclic group could be selected. A more generalized expression would be:

$$v_{n,j} = (j-1)N + (p(n-j)) \,\%\, (N) + 1, \tag{11}$$

for any $p \in \mathbb{N}$ such that $\gcd(p, N) = 1$. For simplicity and clarity, we restrict our analysis to $p = 1$. This construction builds upon the framework of equal-means admissible dice established by Finkelstein and Thorp [4], extending the possible size of non-transitive cycles and relaxing certain constraints on face distributions.

From our explicit formulation, it becomes apparent that as the number of faces $N \to \infty$, the winning probabilities between specific pairs of dice asymptotically approach 1/2. This behavior aligns with the probabilistic findings of Coelho et al. [2], who established a Central Limit Theorem for intransitive dice, demonstrating that the probability of intransitivity often vanishes as the number of faces increases. Our constructive formulas offer a concrete, deterministic perspective that complements their asymptotic probabilistic analysis.

3.1 Properties

A crucial property of this construction is that no face value is repeated, either within a single die or across different dice in the set. This strict uniqueness guarantees that $\mathbb{P}[D_m = D_n] = 0$ for all m, n, completely eliminating the possibility of a tie.

3.1.1 Probability of Dominance

In Appendix B, we derive the analytic expression for the probability that one die in the set rolls higher than another:

$$\mathbb{P}[D_m < D_n] = \frac{1}{2} + \frac{1}{2N} - \frac{(n-m)\,\%\,(N)}{N^2}. \tag{12}$$

A direct consequence of this formula is the translational symmetry: $\mathbb{P}[D_{m+k} < D_{n+k}] = \mathbb{P}[D_m < D_n]$ for all $k = 1, 2, \ldots, N$. This indicates that the probability of dominance depends solely on the cyclic distance between the indices of the dice.

Furthermore, if N is even, then for any pair n, m such that $|n-m| = N/2$, it follows that $\mathbb{P}[D_m < D_n] = \mathbb{P}[D_m > D_n] = 1/2$. Thus, in sets with an even number of dice, every die has exactly one counterpart against which it has a strictly equal chance of winning or losing.

3.1.2 Strong Intransitive Equivalence

Our proposed construction satisfies a strictly stronger version of the strong intransitive property, establishing an equivalence rather than a mere implication.

Theorem For a set of dice $\{D_n\}_{n=1}^{N}$ defined by $v_{n,j} = (j-1)N + (n-j)\,\%\,(N) + 1$, with uniform probabilities $\mathbb{P}[D_n = v_{n,j}] = \frac{1}{N}$, the following equivalence holds:

$$(n-m)\,\%\,(N) < N/2 \iff \mathbb{P}[D_m < D_n] > \frac{1}{2}. \tag{13}$$

Proof Assume $(n-m)\,\%\,(N) < N/2$. This implies:

$$\frac{(n-m)\,\%\,(N)}{N^2} < \frac{1}{2N}. \tag{14}$$

Substituting this inequality into Equation 12 yields the strict lower bound:

$$\mathbb{P}\left[D_m < D_n\right] > \frac{1}{2}. \tag{15}$$

Conversely, assume $\mathbb{P}\left[D_m < D_n\right] > \frac{1}{2}$. Applying Equation 12, we have:

$$\frac{1}{2} + \frac{1}{2N} - \frac{(n-m)\,\%\,(N)}{N^2} > \frac{1}{2}, \tag{16}$$

which simplifies directly to:

$$\frac{N}{2} > (n-m)\,\%\,(N)\,. \tag{17}$$

This completes the proof.

While multiple dice can dominate a given die, this does not readily extend to tournaments with three or more players. For instance, if Player 1 chooses die 1 and Player 2 chooses die $N/2$, there is no third die that can simultaneously beat both. Die 2 beats die 1 but not die $N/2$, and die $N/2+1$ beats die $N/2$ but not die 1. Consequently, this specific generalized set is restricted to two-player tournaments. Expanding this construction to accommodate three or more players remains an open area for future research.

3.2 Examples

The following examples illustrate the structure of the generalized intransitive dice for various values of N.

$N = 3$

$$\begin{array}{ccccccc} D_1 & : & [& 1 & 6 & 8 &] \\ D_2 & : & [& 2 & 4 & 9 &] \\ D_3 & : & [& 3 & 5 & 7 &] \end{array} \tag{18}$$

Utilizing these values on three-sided dice yields the simplest possible set of intransitive dice. Physically, this can be realized by gluing two regular tetrahedra together or by using standard cubic dice and mapping each value to two faces, as seen in standard literature examples.

$N = 4$

$$\begin{array}{cccccccc} D_1 & : & [& 1 & 8 & 11 & 14 &] \\ D_2 & : & [& 2 & 5 & 12 & 15 &] \\ D_3 & : & [& 3 & 6 & 9 & 16 &] \\ D_4 & : & [& 4 & 7 & 10 & 13 &] \end{array} \tag{19}$$

This set is perfectly suited for implementation using regular four-sided dice (tetrahedra), offering an elegant physical representation.

$N = 5$

$$\begin{array}{ccccccccc} D_1 & : & [& 1 & 10 & 14 & 18 & 22 &] \\ D_2 & : & [& 2 & 6 & 15 & 19 & 23 &] \\ D_3 & : & [& 3 & 7 & 11 & 20 & 24 &] \\ D_4 & : & [& 4 & 8 & 12 & 16 & 25 &] \\ D_5 & : & [& 5 & 9 & 13 & 17 & 21 &] \end{array} \tag{20}$$

This is the smallest set where a single die dominates more than one other die. Specifically, $D_5 > \{D_4, D_3\}$, $D_4 > \{D_3, D_2\}$, $D_3 > \{D_2, D_1\}$, $D_2 > \{D_1, D_5\}$, and $D_1 > \{D_5, D_4\}$.

$N = 6$

$$\begin{array}{cccccccccc} D_1 & : & [& 1 & 12 & 17 & 22 & 27 & 32 &] \\ D_2 & : & [& 2 & 7 & 18 & 23 & 28 & 33 &] \\ D_3 & : & [& 3 & 8 & 13 & 24 & 29 & 34 &] \\ D_4 & : & [& 4 & 9 & 14 & 19 & 30 & 35 &] \\ D_5 & : & [& 5 & 10 & 15 & 20 & 25 & 36 &] \\ D_6 & : & [& 6 & 11 & 16 & 21 & 26 & 31 &] \end{array} \tag{21}$$

This configuration maps perfectly onto standard six-sided cubic dice, providing a highly practical and aesthetically pleasing set.

4 Conclusions

This paper has presented a robust, general formula for constructing sets of fair intransitive dice of arbitrary size N, under the constraint that each die possesses exactly N faces. We have derived the explicit analytic expression for the probability of one die dominating another, and utilized this expression to rigorously prove the strong intransitive properties of the generated sets. These dice are ideally suited for two-player *High Roller* games, mathematically guaranteeing that the second player can always select a statistically superior die.

Significant avenues remain for future research. A deeper investigation into the constraints governing the relationship between the number of dice and the number of faces is warranted. This includes verifying whether the proposed constructions, for large N, adhere to the asymptotic limits conjectured by Conrey et al. [3]. Furthermore, generalizing these sets to support *High Roller* tournaments for three or more players remains an open challenge. Achieving this will likely necessitate the relaxation of the unique face value constraint, requiring the repetition of faces within or across dice. This constructive approach should be integrated with advanced probabilistic frameworks, such as the central limit theorems established by Coelho et al. [2], to fully understand the behavior of intransitive dice as complexity scales.

A Probability Derivation for Die Dominance

Consider two independent dice, D_1 and D_2, taking values from sets V_1 and V_2, respectively. We calculate the probability of D_2 rolling strictly higher than D_1 using the law of total probability:

$$\mathbb{P}\left[D_1 < D_2\right] = \sum_{j=1}^{M} \mathbb{P}\left[v_{1,j} < D_2 \mid D_1 = v_{1,j}\right] \mathbb{P}\left[D_1 = v_{1,j}\right]. \tag{22}$$

Given the independence of D_1 and D_2, the conditional probability simplifies:

$$\mathbb{P}\left[v_{1,j} < D_2 \mid D_1 = v_{1,j}\right] = \mathbb{P}\left[v_{1,j} < D_2\right]. \tag{23}$$

Assuming fair dice, $\mathbb{P}\left[D_1 = v_{1,j}\right] = \frac{1}{M}$. Therefore:

$$\mathbb{P}\left[D_1 < D_2\right] = \frac{1}{M} \sum_{j=1}^{M} \mathbb{P}\left[v_{1,j} < D_2\right]. \tag{24}$$

We evaluate $\mathbb{P}\left[v_{1,j} < D_2\right]$ by summing the probabilities of D_2 rolling a value greater than $v_{1,j}$:

$$\mathbb{P}\left[v_{1,j} < D_2\right] = \sum_{k=1}^{M} \mathbb{I}\left[v_{1,j} < v_{2,k}\right] \mathbb{P}\left[D_2 = v_{2,k}\right] = \frac{1}{M} \sum_{k=1}^{M} \mathbb{I}\left[v_{1,j} < v_{2,k}\right]. \tag{25}$$

Substituting this back yields the final expression:

$$\mathbb{P}\left[D_1 < D_2\right] = \frac{1}{M^2} \sum_{j=1}^{M} \sum_{k=1}^{M} \mathbb{I}\left[v_{1,j} < v_{2,k}\right]. \tag{26}$$

Computationally, this involves counting the number of faces on D_2 that exceed each face on D_1, summing these counts, and dividing by M^2.

A.1 Illustrative Example

Consider $N = 3$ and $M = 4$:

$$\begin{array}{lcccccc} D_1: & (& 2 & 2 & 5 & 5 &) \\ D_2: & (& 3 & 3 & 3 & 6 &) \\ D_3: & (& 1 & 4 & 4 & 4 &) \end{array} \tag{27}$$

Applying the formula, we obtain:

$$\mathbb{P}\left[D_1 < D_2\right] = \frac{2+2+2+4}{4^2} = \frac{10}{16} \tag{28}$$

$$\mathbb{P}\left[D_2 < D_3\right] = \frac{0+3+3+3}{4^2} = \frac{9}{16} \tag{29}$$

$$\mathbb{P}\left[D_3 < D_1\right] = \frac{1+1+4+4}{4^2} = \frac{10}{16} \tag{30}$$

B Analytic Probability for the Generalized Set

We apply the formula derived in Appendix A to our specific construction:

$$\mathbb{P}\left[D_m < D_n\right] = \frac{1}{N^2} \sum_{k=1}^{N} \sum_{j=1}^{N} \mathbb{I}\left[v_{m,j} < v_{n,k}\right]. \tag{31}$$

By definition, $v_{n,j} = (j-1)N + (n-j)\,\%\,(N) + 1$. It strictly follows that:

$$(j-1)N < v_{m,j} \leq jN. \tag{32}$$

Therefore, for any m and n, the condition $i < j$ strictly implies $v_{m,i} < v_{n,j}$. We exploit this property to partition the inner sum:

$$\mathbb{P}\left[D_m < D_n\right] = \frac{1}{N^2} \sum_{k=1}^{N} \left[\sum_{j=1}^{k-1} \mathbb{I}\left[v_{m,j} < v_{n,k}\right] + \mathbb{I}\left[v_{m,k} < v_{n,k}\right] + \sum_{j=k+1}^{N} \mathbb{I}\left[v_{m,j} < v_{n,k}\right]\right]. \tag{33}$$

Because $i < j \implies v_{m,i} < v_{n,j}$, we know $\mathbb{I}\left[v_{m,j} < v_{n,k}\right] = 1$ for all $j < k$, and $\mathbb{I}\left[v_{m,j} < v_{n,k}\right] = 0$ for all $j > k$. The equation simplifies to:

$$\begin{aligned}
\mathbb{P}\left[D_m < D_n\right] &= \frac{1}{N^2} \sum_{k=1}^{N} \left[(k-1) + \mathbb{I}\left[v_{m,k} < v_{n,k}\right]\right] \\
&= \frac{1}{N^2} \left[\frac{N(N-1)}{2} + \sum_{k=1}^{N} \mathbb{I}\left[v_{m,k} < v_{n,k}\right]\right].
\end{aligned} \tag{34}$$

We now explicitly evaluate the remaining summation by substituting the definitions of $v_{m,k}$ and $v_{n,k}$:

$$\sum_{k=1}^{N} \mathbb{I}\left[v_{m,k} < v_{n,k}\right] = \sum_{k=1}^{N} \mathbb{I}\left[(m-k)\,\%\,(N) < (n-k)\,\%\,(N)\right]. \tag{35}$$

We analyze this sum by considering two distinct cases, utilizing the property that for $a, b \in \{1, \ldots, N\}$:

$$(a-b)\,\%\,(N) = \begin{cases} a-b & \text{if } b \leq a \\ N+a-b & \text{if } b > a \end{cases} \tag{36}$$

B.1 Case $m < n$

We partition the sum based on the critical points m and n:

$$\begin{aligned}
&\sum_{k=1}^{N} \mathbb{I}\left[(m-k)\,\%\,(N) < (n-k)\,\%\,(N)\right] \\
&\qquad = \sum_{k=1}^{m} \mathbb{I}\left[m < n\right] + \sum_{k=m+1}^{n} \mathbb{I}\left[N + m < n\right] + \sum_{k=n+1}^{N} \mathbb{I}\left[m < n\right] \\
&\qquad = m + 0 + (N - n) = N + m - n.
\end{aligned} \tag{37}$$

B.2 Case $n < m$

Similarly, partitioning the sum yields:

$$\begin{aligned}
&\sum_{k=1}^{N} \mathbb{I}\left[(m-k)\,\%\,(N) < (n-k)\,\%\,(N)\right] \\
&\qquad = \sum_{k=1}^{n} \mathbb{I}\left[m < n\right] + \sum_{k=n+1}^{m} \mathbb{I}\left[m < N + n\right] + \sum_{k=m+1}^{N} \mathbb{I}\left[m < n\right] \\
&\qquad = 0 + (m - n) + 0 = m - n.
\end{aligned} \tag{38}$$

Combining both cases, we establish a unified expression:

$$\sum_{k=1}^{N} \mathbb{I}\left[(m-k)\,\%\,(N) < (n-k)\,\%\,(N)\right] = \begin{cases} m - n & \text{if } n < m \\ N + m - n & \text{if } n > m \end{cases} \tag{39}$$

Given that $n \neq m$, this precisely matches the definition of the modulo operator:

$$\sum_{k=1}^{N} \mathbb{I}\left[(m-k)\,\%\,(N) < (n-k)\,\%\,(N)\right] = (m-n)\,\%\,(N)\,. \tag{40}$$

Substituting this result back into the probability equation provides:

$$\mathbb{P}\left[D_m < D_n\right] = \frac{1}{N^2}\left[\frac{N(N-1)}{2} + (m-n)\,\%\,(N)\right]. \tag{41}$$

Using the identity $(m-n)\,\%\,(N) = N - (n-m)\,\%\,(N)$, we arrive at the final analytic form:

$$\begin{aligned}
\mathbb{P}\left[D_m < D_n\right] &= \frac{1}{N^2}\left[\frac{N(N+1)}{2} - (n-m)\,\%\,(N)\right] \\
&= \frac{1}{2} + \frac{1}{2N} - \frac{(n-m)\,\%\,(N)}{N^2}.
\end{aligned} \tag{42}$$

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